

INVITED PAPER

ON MEASURING THE AMOUNT OF INFORMATION FROM
THE DATA IN A BAYESIAN ANALYSIS
(WITH DISCUSSION)

D.J. De Waal: University of the Orange Free State, P.O. Box 339,
Bloemfontein, 9300.
and
P.C.N. Groenewald:

Keywords: Entropy, expected information, information theory.

Summary: The amount of information on a parameter θ that is provided by our data $x_1, x_2, \dots, x_n$ is stochastic and is worthwhile investigating from a Bayesian point of view. It is assumed that the model $f(x|\theta)$ is fixed and that we have some prior knowledge $\pi(\theta)$ available on θ . The measure introduced by Shannon (1948) is considered and examined in some detail in a life testing problem. The case of more than one parameter such as the normal for instance, is also considered. The effect of the prior on the amount of information is also discussed for the normal and binomial cases.

1. INTRODUCTION

There are various aspects and measures that could be considered when one thinks of information from data. The whole concept of information theory in the exact sense of Shannon (1948) which has a long history, seems to be appropriate when combined with Bayesian analysis. There is a large selection of literature available on information theory such as the works of Hartley (1928), Shannon (1948), Feinstein (1958), Good (1950, 1960), Kullback (1959), Renyi (1960), Osteyee and Good (1974) and Aczel and Daroczy (1975), to mention a few. However, it is only since Lindley (1956) that the idea of information from the results of a statistical experiment has been stressed. Our aim with this paper is to deal specifically with the amount of information obtainable from data on an unknown parameter Θ , from a Bayesian point of view. By discussing this type of information through a few examples we hope to bring to the attention of statisticians the concept of the amount of information actually received from data compared to what could have been obtained. We assume an unknown parameter θ belonging to a parameter space Θ and we want to know more about θ . Let $\pi(\theta)$ be a prior probability density of θ with respect to some dominating measure on Θ . Let $\epsilon = \{S_X, \Theta, f(x)|\theta\}$ be the experiment comprising one observation x of the random quantity $X \in S_X$ which is distributed, for some $\theta \in \Theta$, according to the probability density $f(x|\theta)$ with respect to some σ -finite dominating measure on S_X . Without loss of generality, we shall assume that the probability densities of X which correspond to different values of θ differ at least on a set of non-zero measure.

Let the data consist of n independent observations $x_1, x_2, \dots, x_n$ of the random variable X . The question we would like to consider here is: "How much information on θ did we obtain from this data?" To answer this question, we first have to find some measure for this information. There are several measures of information that were proposed in the literature such as the Kullback-Leibler and Fisher information statistics. The variance of the posterior probability density, $\pi(\theta|x_1, \dots, x_n)$, may also be regarded as a measure. The measure that we will be considering here, because

of preferable properties, was introduced by Shannon (1948) and further investigated by Lindley (1956) who defined it as:

Definition 1

The amount of information provided by the prior $\pi(\theta)$ is

$$(1.1) \quad \begin{aligned} I_{\theta} &= E_{\theta} \log \pi(\theta) \\ &= \int_{\Theta} \pi(\theta) \log \pi(\theta) d\theta. \end{aligned}$$

For any θ for which $\pi(\theta) = 0$, define $\pi(\theta) \log \pi(\theta)$ to be zero. E_Y denotes the expected value taken with respect to the marginal density of Y .

The negative of I_{θ} is sometimes called the entropy of the distribution π and in a sense measures the amount of uncertainty inherent in the probability distribution.

Definition 2

The amount of information provided by the experiment ϵ with prior knowledge $\pi(\theta)$, when the observation is x , is

$$(1.2) \quad I_x = I_{\theta|x} - I_{\theta}$$

where

$$I_{\theta|x} = \int_{\Theta} \pi(\theta|x) \log \pi(\theta|x) d\theta.$$

Define the integral to be zero if $\pi(\theta|x) = 0$.

It has been pointed out by Lindley that the measures (1.1) and (1.2) are not invariant under a change of description of the parameter space. The quantity $I_{\theta|x}$ depends on x and measures the amount of information provided by the posterior. The fact that I_{θ} , the amount of information on θ provided by the prior, has been removed, leaves us with the information from the data. We will therefore consider I_x in some detail and in fact suggest the use of I_x as the measure to use in experiments.

Definition 3

The average amount of information provided by the experiment ϵ , with prior knowledge $\pi(\theta)$, is

$$(1.3) \quad I(\epsilon, \pi(\theta)) = E_X(I_X).$$

It has been pointed out by Lindley (1956) that an alternative expression for $I(\epsilon, \pi(\theta))$ is

$$(1.4) \quad I(\epsilon, \pi(\theta)) = \int_S \int_{\Theta} g(x, \theta) \log \left\{ \frac{g(x, \theta)}{m(x)\pi(\theta)} \right\} d\theta dx$$

where $g(x, \theta) = f(x|\theta)\pi(\theta)$ is the joint density of X and θ and

$$m(x) = \int_{\Theta} g(x, \theta) d\theta$$

is the marginal density of X .

Expression (1.4) shows the symmetry between x and θ and also shows the fact that $I(\epsilon, \pi(\theta))$ is invariant under a one-to-one transformation of the parameter space.

The average amount of information provided by $I(\epsilon, \pi(\theta))$ is quite useful as a point of reference for I_x , since I_x is often unbounded when S_X and Θ are continuous and unbounded. Absolute certainty about the value of the parameter is then implied only when $I_x = \infty$.

Lindley (1956) mentioned in his paper that $I(\epsilon, \pi(\theta))$ can be used to compare two experiments ϵ_1 and ϵ_2 with each other and defined ϵ_1 to be more informative than ϵ_2 if $I(\epsilon_1, \pi(\theta)) > I(\epsilon_2, \pi(\theta))$. Analogously we will state the following definition on I_x :

Definition 4

Let ϵ_1 and ϵ_2 be two experiments resulting in observing x and y respectively from the same distribution $f(.|\theta)$ with the same prior density $\pi(\theta)$ for both experiments, then x is more informative than y if

$$I_x > I_y.$$

If $I_x = I_y$ then x and y are equally informative.

Here x and y can be data sets consisting of the same number of observations n . If the sample sizes of the two experiments differ, then I_x and I_y are not directly comparable. Bernardo (1979a) used the measure $I(\epsilon, \pi(\theta))$ to suggest an operative procedure to derive reference posterior distributions which approximately describe the kind of inferences which one is entitled to make with little relevant initial information. In contrast, we will not consider any formulation of priors or posteriors, but assume that the prior information is available. Bernardo (1981) also considers the information from a decision theoretic point of view in establishing reference priors. The following properties of $I(\epsilon, \pi(\theta))$ have been proved by Lindley (1956).

Properties

p 1. $I(\epsilon, \pi(\theta)) \geq 0$ with equality if $f(x|\theta)$ does not depend θ , except possibly in a null set for θ .

p 2. If ϵ_1 and ϵ_2 are two consecutive experiments resulting in the experiment ϵ , then

$$(1.5) \quad I(\epsilon_1, \pi(\theta)) + I(\epsilon_2, \pi(\theta)) = I(\epsilon, \pi(\theta)).$$

This property can be used to determine a sequential sampling procedure if expected information is treated as expected utilities (Lindley (1971), Bernardo (1979b)). Before averaging over the sample space, (1.5) can be written as follows:

$$(1.6) \quad I_{x_1} + I_{x_2|x_1} = I_{x_1, x_2}$$

where

$$(1.7) \quad I_{x_2|x_1} = E_{\theta|x_1, x_2} \log \pi(\theta|x_1, x_2) - E_{\theta|x_1} \log \pi(\theta|x_1)$$

and

$$(1.8) \quad I_{x_1, x_2} = E_{\theta|x_1, x_2} \log \pi(\theta|x_1, x_2) - I_{\theta}.$$

Comparing $I_{x_2|x_1}$ with I_{x_1} and I_{x_1, x_2} one can determine the gain in information provided by the additional observation.

p 3. For fixed ϵ , $I(\epsilon, \pi(\theta))$ is a concave function of $\pi(\theta)$.

Various other properties of $I(\epsilon, \pi(\theta))$ are given by Lindley (1956) but for our purposes the three given above will suffice.

Before considering some specific cases, we would like to mention some other measures of information.

Kullback-Leibler measure

The Kullback-Leibler (1951) measure of divergence between the prior and posterior distributions is defined as

$$(1.9) \quad J_x = E_{\theta|x} \log \left\{ \frac{\pi(\theta|x)}{\pi(\theta)} \right\}.$$

It can be shown that $E_X(J_X)$ and (1.3) are identical. Thus the average amount of information provided by an experiment is the expected value over all $x \in S_x$ of the KL-measure of divergence between the posterior and prior distributions. Although

$$E_X(J_X) = I(\epsilon, \pi(\theta)) = E_X(I_{\theta|X}) - I_{\theta}$$

it does not mean that $I_{\theta|X} - I_{\theta}$ is equal to J_X since

$$J_X = I_{\theta|X} - E_{\theta|X} \log \pi(\theta),$$

while

$$I_{\theta|X} - I_{\theta} = I_{\theta|X} - E_{\theta} \log \pi(\theta).$$

Fisher information statistic

The wellknown Fisher information statistic

$$(1.10) \quad i(\theta) = E_{X|\theta} \left\{ \frac{\partial \log f(X|\theta)}{\partial \theta} \right\}^2$$

gives the extent to which uncertainty regarding a specific unknown value of θ is reduced after averaging over all possible values of X .

According to Rao (1973) p 332, $i(\theta)$ is a measure of sensitivity of the random variable X with respect to an infinitesimal change in the value of θ . In asymptotic theory (see de Groot (1970) p.213) $i(\hat{\theta})$, where $\hat{\theta}$ is the maximum likelihood estimate for θ , gives the precision (inverse of the variance) of the posterior distribution of θ . As a measure on the variance of the posterior, it certainly can measure information from the data, but then the actual posterior variance can be used instead. Therefore $i(\theta)$ does not seem to be especially useful as a measure of information on θ for a specific set of data. We will first consider the use of I_x in a life testing example. In section 3 the case of more than one parameter such as the normal will be discussed, while the sensitivity of I_x with respect to prior specifications is examined in section 4.

2. THE BEHAVIOUR OF I_x IN A LIFE TESTING PROBLEM

Berger and O'Hagan (1987) consider the life testing example given in Martz and Waller (1982) to show features of robustness of prior information. We will use the same example to investigate the behaviour of I_x .

Let θ be the mean life of a new industrial engine. Data become available in the form of two independent life-times that have been observed, namely, $x_1 = 2000$ and $x_2 = 2500$. We want to know just how much information x_1 and x_2 provide on θ . The mathematical formulation of the problem is quite straightforward. Let the random variable X conditional on θ be exponentially distributed with density function given by

$$(2.1) \quad f(x|\theta) = \frac{1}{\theta} \exp(-x/\theta), x > 0.$$

It is clear that $E(X|\theta) = \theta$. The belief of an engineer about θ is expressed in terms of the following probabilities.

TABLE 1
PRIOR PROBABILITIES OF THE ENGINEER ON θ

θ	[0, 1000)	[1000, 2000)	[2000, 3000)	[3000, 4000)	[4000, 5000)
$\pi_i = \int \pi(\theta) d\theta$	0,01	0,04	0,20	0,50	0,15
θ	[5000, 6000)	[6000, 7000)	[7000, 8000)	[8000, 9000)	[9000, 10000)
π_i	0,05	0,03	0,01	0,008	0,002

Our prior density $\pi(\theta)$ is in the form of a histogram, namely

$$(2.2) \quad \pi(\theta) = \pi_i/1000 = \pi_i^* \text{ say } , \theta \in \mathcal{I}_i$$

where $\mathcal{I}_i$ indicates the i -th interval. Note that $E_\theta(\theta) = 3622$ so that the two observations with a mean $\bar{x} = 2250$ may indicate some different information than that of the engineer. Applying (1.1) to $\pi(\theta)$ we obtain

$$I_\theta = -8,3876.$$

This is the amount of information on θ that we have from the prior. To calculate I_{x_1, x_2} we need the posterior density $\pi(\theta|x_1, x_2)$. This is given by

$$(2.3) \quad \begin{aligned} \pi(\theta|x_1, x_2) &= \frac{\pi(\theta) \exp(-(x_1 + x_2)/\theta)}{\theta^2 m(x_1, x_2)}, \quad 0 < \theta < \infty \\ &= \frac{\pi_i^* \exp(-(x_1 + x_2)/\theta)}{\theta^2 m(x_1, x_2)}, \quad \theta \in \mathcal{I}_i \end{aligned}$$

where the marginal density of X_1 and X_2 is given by

$$(2.4) \quad m(x_1, x_2) = \sum_{i=1}^{10} \frac{\pi_i^*}{(x_1 + x_2)} \left\{ e^{-(x_1 + x_2)/b_i} - e^{-(x_1 + x_2)/a_i} \right\},$$

with

$$[a_i, b_i) = \mathcal{I}_i.$$

The associated posterior probability $\pi_i(x_1, x_2)$ for interval I_i can be calculated from (2.3), namely

$$(2.5) \quad \pi_i(x_1, x_2) = P(\theta \in I_i | x_1, x_2) \\ = \frac{\pi_i^*}{m(x_1, x_2)(x_1 + x_2)} \left\{ e^{-(x_1+x_2)/b_i} - e^{-(x_1+x_2)/a_i} \right\}$$

and are given in the table 2 for $x_1 = 2000$ and $x_2 = 2250$ where $m(2000, 2500) = 2,137 \times 10^{-8}$.

TABLE 2
POSTERIOR PROBABILITIES GIVEN $x_1 = 2000, x_2 = 2250$.

I_i $\pi_i(x_1, x_2)$	I_1 0,0012	I_2 0,0039	I_3 0,2448	I_4 0,5278	I_5 0,1278
I_i $\pi_i(x_1, x_2)$	I_6 0,0342	I_7 0,0167	I_8 0,0046	I_9 0,0032	I_{10} 0,0007

If we approximate $\log \pi(\Theta/x_1, x_2)$ by $\log(\pi_i(x_1, x_2)/1000)$, $\theta \in I_i$, then we get the following approximate value:

$$I_{\theta|2000,2500} \approx -8,2181.$$

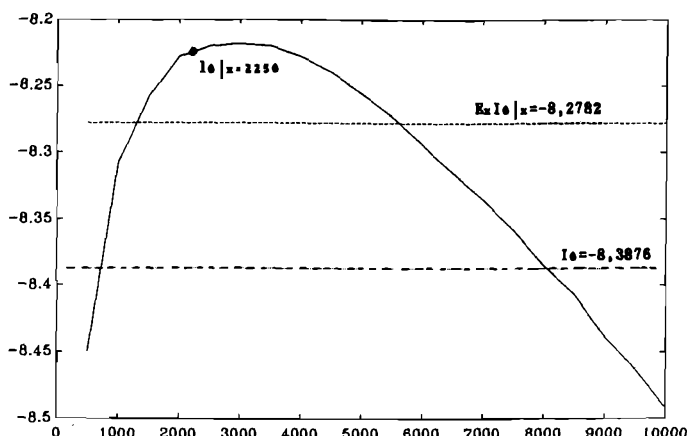
Comparing this with $I_{\theta} = -8,3876$, we can see that the observations have increased the information on θ .

The amount of information obtained from the data according to (1.2), is

$$I_{(2000,2500)} = -8,2181 + 8,3876 \\ = 0,1695$$

To give any meaning to this value, we need to compare it with $I_{(x_1, x_2)}$ for all other x_1, x_2 values. The values of $I_{\theta|x}$ have been calculated for $500 < \bar{x} < 10000$ and plotted in figure 1.

Figure 1: $I_\theta, I_{\theta|\bar{x}}$ and $E_{\bar{x}} I_{\theta|\bar{x}}$ as a function of $\bar{x}$ when $n = 2$ in the life testing example.



The maximum of $I_{\theta|\bar{x}}$ in this interval occurs at $\bar{x} = 3000$ approximately, which means that the amount of information from the observed values ($\bar{x} = 2250$) is not far from the maximum amount of information possible on θ . From figure 1 it can be seen that for $500 < \bar{x} < 8000$, approximately, we will gain information from the data since it supports the prior information. Otherwise the information becomes contradictory to our prior information. For more extreme observations, see figure 3. The average amount of information $I(\epsilon, \pi(\theta))$ given in (1.3) involves complicated integration to calculate exactly, but by approximating the graph of $I_{\theta|\bar{x}}$ in figure 1 by a step function in the interval T_i , it can be calculated approximately.

Let $Z = X_1 + X_2$, then the density of Z conditional on θ is given by

$$g(z|\theta) = \frac{z}{\theta^2} e^{-z/\theta}, z > 0.$$

Therefore

$$\begin{aligned} P(z_1 \leq Z < z_2|\theta) &= \frac{1}{\theta^2} \int_{z_1}^{z_2} z e^{-z/\theta} dz \\ &= P(\theta|z_1, z_2), \text{ say.} \end{aligned}$$

Unconditionally, therefore

$$\begin{aligned}
 (2.6) \quad P(z_1 < Z < z_2) &= \sum_{i=1}^{10} \pi_i^* \int_{z_i} P(\theta|z_1, z_2) d\theta \\
 &= \sum_{i=1}^{10} \pi_i^* \left\{ -b_i (e^{-z_2/b_i} - e^{-z_1/b_i}) \right. \\
 &\quad \left. + a_i (e^{-z_2/a_i} - e^{-z_1/a_i}) \right\}.
 \end{aligned}$$

The following marginal probabilities were calculated for $\bar{X} = \frac{1}{2}(X_1 + X_2)$ and the associated $I_{\theta|\bar{x}}$ values obtained from figure 1:

TABLE 3
MARGINAL PROBABILITIES FOR $\bar{X} = \frac{1}{2}(X_1 + X_2)$
AND ASSOCIATED $I_{\theta|\bar{x}}$ VALUES

$\bar{x}$	$\int m(\bar{x})d\bar{x}$	$I_{\theta \bar{x}}$	$\bar{x}$	$\int m(\bar{x})d\bar{x}$	$I_{\theta \bar{x}}$
[0, 500)	0,0468	-8,455	[5000, 5500)	0,0388	-8,260
[500, 1000)	0,0894	-8,332	[5500, 6000)	0,0324	-8,280
[1000, 1500)	0,1035	-8,265	[6000, 6500)	0,0270	-8,305
[1500, 2000)	0,1032	-8,235	[6500, 7000)	0,0220	-8,326
[2000, 2500)	0,0961	-8,216	[7000, 7500)	0,0178	-8,350
[2500, 3000)	0,0860	-8,212	[7500, 8000)	0,0155	-8,376
[3000, 3500)	0,0751	-8,213	[8000, 8500)	0,0129	-8,405
[3500, 4000)	0,0645	-8,220	[8500, 9000)	0,0107	-8,430
[4000, 4500)	0,0549	-8,230	[9000, ∞)	0,0571	-8,475
[4500, 5000)	0,0463	-8,245			

$I_{\bar{x}}$ can be found from

$$I_{\bar{x}} = I_{\theta|\bar{x}} - I_{\theta}.$$

From table 3 follows that

$$\begin{aligned}
 I(\epsilon, \pi(\theta)) &\simeq \Sigma I_{\theta|\bar{x}} \int m(\bar{x})d\bar{x} - I_{\theta} \\
 &= -8,2782 + 8,3876 \\
 &= 0,1094.
 \end{aligned}$$

The amount of information obtained from $x_1 = 2000$ and $x_2 = 2500$ is substantially higher than the expected amount of information for two observations.

The question arises as to what happens to I_x if one more observation is taken. We will now consider this.

Taking the observations sequentially.

Let us consider the behaviour of I_{x_1} after only $x_1 = 2000$ has been observed. For this case the posterior probabilities have been obtained through numerical integration. The posterior density is given by

$$\begin{aligned}\pi(\theta|x_1) &= \frac{\pi(\theta)}{\theta m(x_1)} e^{-x_1|\theta}, \quad \theta > 1 \\ &= \frac{\pi_i^*}{\theta m(x_1)} e^{-x_1|\theta}, \quad \theta \in I_i\end{aligned}$$

TABLE 4
POSTERIOR PROBABILITIES FOR $x_1 = 2000$

I_i $\int_{I_i} \pi(\theta 2000)d\theta$	I_1	I_2	I_3	I_4	I_5
	0,0031	0,0435	0,2283	0,5144	0,1364
I_i $\int_{I_i} \pi(\theta 2000)d\theta$	I_6	I_7	I_8	I_9	I_{10}
	0,0404	0,0217	0,0065	0,0047	0,0011

From table 4 follows that

$$I_{2000} = -8,2910 + 8,3876 = 0,0966.$$

Using (1.6), it follows that the additional amount of information gained by taking $x_2 = 2500$ is

$$\begin{aligned}I_{x_2|x_1} &= I_{x_1, x_2} - I_{x_1} \\ &= 0,1667\end{aligned}$$

This is almost twice that gained from $x_1 = 2000$ alone.

Figure 2: $I_{\bar{x}}$ is a function of $\bar{x}$ for $n = 1, 2, 3, 4, 5$.

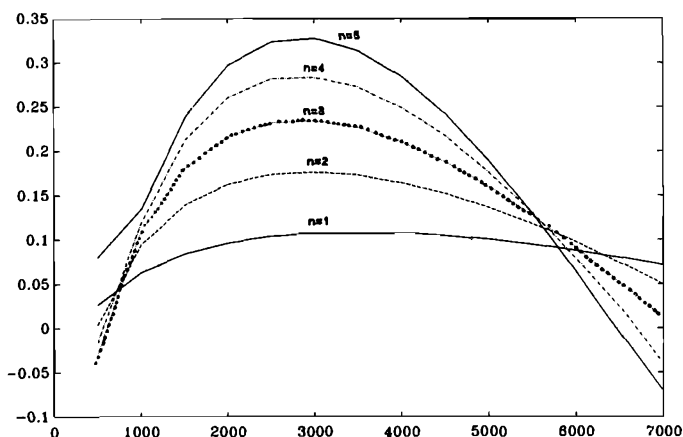


Figure 2 shows $I_{\bar{x}}$ for $n = 1, 2, 3, 4, 5$ and it is clear that $I_{\bar{x}}$ only increases with n when $\bar{x}$ lies in an interval from about 1000 to 5000. This is a rather interesting result for it means that, for instance, 5 observations with a mean of $\bar{x} = 7000$ give negative information and it seems that an increase in n leads to a decrease in I_{7000} . An explanation for this is that the information from the data becomes more and more opposed to the prior information, so that there is more uncertainty about the parameter after the experiment than before.

In the range where $I_{\bar{x}}$ is positive, we notice that three good observations can give more information than five bad ones. By good observation is meant observations that lie close to the maximum of the curve. It is clear of course, that the information increases as n increases. In fact, as Lindley also pointed out, it tends to infinity for all $\bar{x}$ as n tends to infinity.

When investigating the behaviour of $I_{\bar{x}}$ for $n > 5$ and for $\bar{x} < 1000$ and $\bar{x} > 7000$, the numerical integrations come under suspicion and therefore a discrete approximation to the problem is done. This seems to work quite well and is a procedure we would suggest in general in order to obtain an approximate profile of $I_{\bar{x}}$.

Let θ_i be the midpoint of $\theta \in \mathcal{I}_i$, then the posterior probabilities, given $x_1, \dots, x_n$, are approximated by

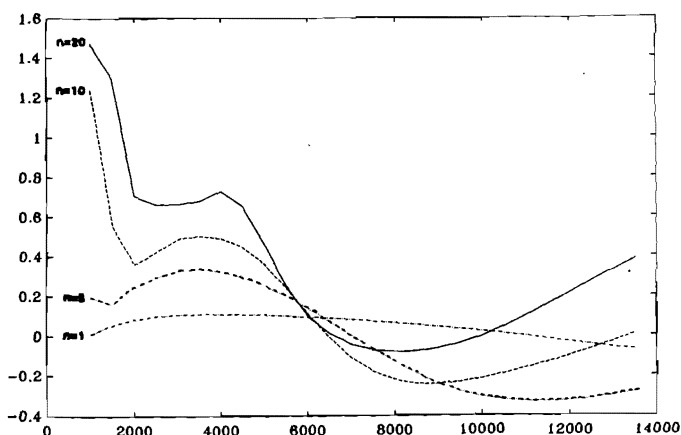
$$\pi(\theta_i|\bar{x}) = \frac{\pi_i \theta_i^{-n} \exp(-n\bar{x}/\theta_i)}{\sum \pi_i \theta_i^{-n} \exp(-n\bar{x}/\theta_i)}, \quad i = 1, \dots, 10.$$

Therefore

$$I_{\bar{x}} \simeq \sum_{i=1}^{10} \pi(\theta_i|\bar{x}) \log \pi(\theta_i|\bar{x}) - \sum_{i=1}^{10} \pi_i \log \pi_i.$$

These profiles of $I_{\bar{x}}$ have been graphed in figure 3 for $n = 1, 5, 10, 20$.

Figure 3: Profiles of $I_{\bar{x}}$ for $n = 1, 5, 10, 20$ for life testing example.



From figure 3 can be seen that after an initial decrease of information from the sample as the data starts to contradict the prior, the information starts to increase again with extreme values of $\bar{x}$. This kind of profile seems common with bounded parameter spaces (see section 4) and the interpretation of the high information at extreme observations should be that something is wrong with the data, the prior or the model. In figure 3, that would be when a mean outside the interval (2000, 8000) is obtained when $n = 20$.

3. INFORMATION ON MORE THAN ONE PARAMETER: THE NORMAL CASE

Extending Lindley's (1956) definition to the case of two parameters we define:

Definition 5

The amount of information on both θ and γ provided by the joint prior $\pi(\theta, \gamma)$ is given by

$$(3.1) \quad \begin{aligned} I_{\theta, \gamma} &= \int_{\Theta} \int_{\Gamma} \pi(\theta, \gamma) \log \pi(\theta, \gamma) d\theta d\gamma \\ &= I_{\theta} + E_{\theta} I_{\gamma|\theta}. \end{aligned}$$

Definition 6

The amount of information provided by the experiment ϵ on θ and γ , with prior knowledge $\pi(\theta, \gamma)$, when the observation is x , is given by

$$(3.2) \quad I_x^{\theta, \gamma} = I_{\theta, \gamma|x} - I_{\theta, \gamma}$$

where

$$I_{\theta, \gamma|x} = I_{\theta|x} + E_{\theta|x} I_{\gamma|\theta, x}.$$

These definitions can easily be extended to any number of parameters. The amount of information provided by the experiment on θ alone is given by

$$I_x^{\theta} = I_{\theta|x} - I_{\theta}$$

as in (1.2), but note that the expected amount of information on θ ,

$$I^{\theta}(\epsilon, \pi(\theta, \gamma)) = E_X(I_X^{\theta})$$

depends on the entire prior, $\pi(\theta, \gamma)$, since the expectation is with respect to the marginal distribution of X , and

$$m(x) = \int_{\Theta} \int_{\Gamma} f(x|\theta, \gamma) \pi(\theta, \gamma) d\theta d\gamma.$$

$I_{\gamma|\theta}$ can be interpreted as the information on γ from the prior conditionally on θ known, and (3.1) can also be written as

$$I_{\theta, \gamma} = I_{\gamma} + E_{\gamma} I_{\theta|\gamma}.$$

Consider now the normal-gamma case where X , given m and r , is normally distributed with mean m and precision r . The prior on M , given r , is normal with mean μ and precision τr while R has a gamma distribution with parameters α and β . So if X represents a random sample of size n , then

$$f(x|m, r) = \prod_{i=1}^n n(x_i|m, r^{-1})$$

where

$$(3.3) \quad M|r \sim N(\mu, (\tau r)^{-1}),$$

$$R \sim \text{Gamma}(\alpha, \beta).$$

Then, from (3.1), the information from the prior is given by

$$I_{r,m} = I_r + E_r I_{m|r}$$

where

$$(3.4) \quad I_r = \log \beta - \log \Gamma(\alpha) + (\alpha - 1)\Psi(\alpha) - \alpha,$$

$$(3.5) \quad E_r I_{m|r} = \frac{1}{2} \left(\frac{\tau}{2\pi\beta e} \right) + \frac{1}{2} \Psi(\alpha)$$

and

$$\Psi(\alpha) = \frac{d}{d\alpha} \log \Gamma(\alpha).$$

After the data has been observed, the posteriors are

$$M|r, X \sim N\left(\mu', \frac{1}{(\tau + x)r}\right)$$

and

$$R|X \sim \text{Gamma}(\alpha', \beta')$$

where

$$\mu' = \frac{\tau\mu + n\bar{x}}{\tau + n},$$

$$\alpha' = \alpha + \frac{n}{2},$$

$$\beta' = \beta + \frac{1}{2}S + \frac{\tau n(\bar{x} - \mu)}{2(\tau + n)}$$

and

$$S = \sum_{i=1}^n (x_i - \bar{x})^2.$$

The information provided by the data is then

$$\begin{aligned} I_x^{r,m} &= (I_{r|x} - I_r) + (E_{r|x} I_{m|r,x} - E_r I_{m|r}) \\ &= I_x^r + I_x^{m|r} \end{aligned}$$

where the first term on the right-hand side is I_x^r , the information on R alone, while the second represents the additional information provided on M .

It follows that

$$\begin{aligned} (3.6) \quad I_x^r &= \log \frac{\beta'}{\beta} - \log \frac{\Gamma(\alpha')}{\Gamma(\alpha)} + (\alpha' - 1)\Psi(\alpha') - (\alpha - 1)\Psi(\alpha) - \frac{n}{2} \\ &= \log \left(1 + \frac{1}{2} \frac{S}{\beta} + \frac{\tau n (\bar{x} - \mu)^2}{2(\tau + n)\beta} \right) - \log \prod_{k=\alpha}^{\alpha + \frac{n}{2} - 1} k \\ &\quad + (\alpha - 1) \sum_{k=\alpha}^{\alpha + \frac{n}{2} - 1} \frac{1}{k} + \frac{n}{2} \sum_{k=1}^{\alpha + \frac{n}{2} - 1} \frac{1}{k} - \frac{n}{2}(C + 1) \end{aligned}$$

when α is a positive integer, n is even and $C = 0.577215$ is Euler's constant. The information on M is given by

$$\begin{aligned} (3.7) \quad I_x^{m|r} &= \frac{1}{2} \log \left(\frac{(\tau + n)\beta}{\tau\beta'} \right) + \frac{1}{2} (\Psi(\alpha') - \Psi(\alpha)) \\ &= -\frac{1}{2} \log \left(1 + \frac{1}{2} \frac{S}{\beta} + \frac{\tau n (\bar{x} - \mu)^2}{2(\tau + n)\beta} \right) + \frac{1}{2} \sum_{k=\alpha}^{\alpha + \frac{n}{2} - 1} \frac{1}{k} \\ &\quad + \log \left(1 + \frac{n}{\tau} \right) \end{aligned}$$

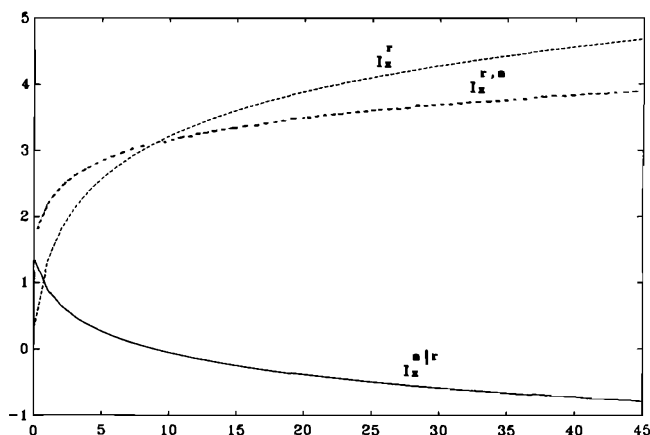
for n even.

From (3.6) and (3.7) it can be seen that, for fixed priors, the information provided by the data on R increases with the sample variance, $\frac{S}{n}$, while the information on M decreases with $\frac{S}{n}$. This is expected, since a large sample variance implies little information on the mean, but substantial information on the inverse of the variance, R .

The squared deviation $(\bar{x} - \mu)^2$ has a similar effect since it gives information on the mean as well as the precision due to the construction of the model in (3.3).

It is interesting to note the effect that the prior on R has on the information provided by the data. In figure 4 I_x^r , $I_x^{m|r}$ and the sum $I_x^{r,m}$ are plotted as functions of $\frac{S}{n}$ for $n = 10$, $\tau = \frac{1}{2}$, $(\bar{x} - \mu)^2 = 4$ and a "weak" prior on R , namely $\alpha = 40$ and $\beta = 2$.

Figure 4: Profiles of $I_x^{r,m}$, I_x^r and $I_x^{m|r}$ as a function of $\frac{S}{n}$ when $(\bar{x} - \mu)^2 = 4$, $\tau = \frac{1}{2}$, $n = 10$ and $\alpha = 40$, $\beta = 2$.



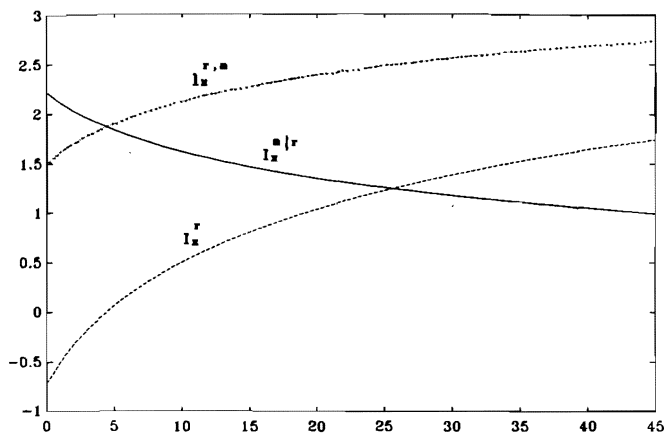
In Figure 5 a "strong" prior ($\alpha = 2$, $\beta = 20$) is used. This reduces the amount of information on R obtainable from the data, but increases the amount of information on M since the resulting prior on M is now weaker.

The total amount of information, $I_x^{r,m}$ is a function of the sufficient statistics $\bar{X}$ and S , and the expected amount of information,

$$I^{m,r}(\epsilon, \pi(m, r)) = E_X(I_X^{m,r}),$$

can be obtained by taking expectations with respect to the joint marginal of $\bar{X}$ and S ,

Figure 5: Profiles of I_x^r, I_x^r and $I_x^{m|r}$ as a function on $\frac{s}{n}$ when $(\bar{x} - \mu)^2 = 4, \tau = \frac{1}{2}, n = 10$ and $\alpha = 2, \beta = 20$.



$$\begin{aligned}
 (3.8) \quad m(\bar{x}, s) &= \frac{\beta^\alpha \Gamma(\alpha + \frac{n}{2})}{2^{\frac{n}{2}} \pi^{\frac{1}{2}} \Gamma(\alpha) \Gamma(\frac{n-1}{2})} \left(\frac{n\tau}{n+\tau} \right)^{\frac{1}{2}} S^{\frac{n-3}{2}} \\
 &\quad \left[\beta + \frac{1}{2} S + \frac{n\tau(\bar{x} - \mu)^2}{2(n+\tau)} \right]^{-(\alpha + \frac{n}{2})} \\
 &= m_1(\bar{x}|S) m_2(S),
 \end{aligned}$$

where $\bar{X}|S$ has a generalized t-distributions with parameters $2\alpha + n - 1$ and $\frac{n\tau(2\alpha + n - 1)}{(2\beta + S)(n + \tau)}$, and $\frac{\alpha}{\beta(n-1)}S$ has a F-distribution with $n - 1$ and 2α degrees of freedom. The expectation must however be evaluated numerically.

4. EFFECT OF THE PRIOR ON I_x

In this section we will show through two examples how the choice of the prior can affect the profile of the information statistic I_x . We will not go into the matter of choosing the best prior to represent vague prior information. This has been done by Bernardo (1979a), (1981), who defines a vague prior as the prior with respect to which the expected information in the data is a maximum. The important point is that the more vague the prior is, the more the additional information that can be obtained from the data. This comes out clearly in example 3, where I_x increases as the prior gets more vague.

We assume that prior information is available in the form given in the life testing example in table 1 or that the information is expressed in terms of a wellknown distribution. The two cases that will be treated here are the normal and the binomial models.

Example 2. (The normal model)

Assume that the random variable X is normally distributed with unknown mean θ and known variance $\sigma^2 = 2$. It is common practice in Bayesian analysis to assume that θ is normally distributed with mean μ and variance τ where μ and τ are obtained from prior knowledge. Let us assume that our prior knowledge on θ is that $\mu = 10$ and $\tau = 0,4788$. Lindley (1956) has shown that the amount of information from a sample of n observations with mean $\bar{x}$ stays constant for all possible $\bar{x}$, namely

$$(4.1) \quad I_{\bar{x}} = \frac{1}{2} \log(1 + n\tau^2/\sigma^2).$$

For the example with say $n = 3$ observations,

$$I_{\bar{x}} = 0,2706.$$

This means that the information from the data stays constant at 0,2706 independent of what the observations are. This is however true only under the assumption of normality for the prior distribution on θ . Suppose instead that our prior information was given in terms of the following probabilities:

TABLE 5
PRIOR PROBABILITIES π_i FOR θ

θ	7 - 7,5	7,5 - 8	8 - 8,5	8,5 - 9	9 - 9,5	9,5 - 10
π_i	0,002	0,008	0,01	0,05	0,1	0,3
θ	10 - 10,5	10,5 - 11	11 - 11,5	11,5 - 12	12 - 12,5	12,5 - 13
π_i	0,3	0,1	0,05	0,01	0,008	0,002

For the above distribution $\mu = 10, \tau = 0,4788$ and the third and fourth moments about the mean are respectively $E(\theta - 10)^3 = 0, E(\theta - 10)^4 = 1,1362$. The normal prior does not fit this information because the kurtosis of the above distribution is 4,956 compared with 3 for the normal.

The behaviour of $I_{\bar{x}}$ for this prior can be approximated by assuming θ to be discrete taking on the midvalues of the various intervals. Then $I_{\bar{x}}$ can be calculated as follows: The posterior probabilities are approximately

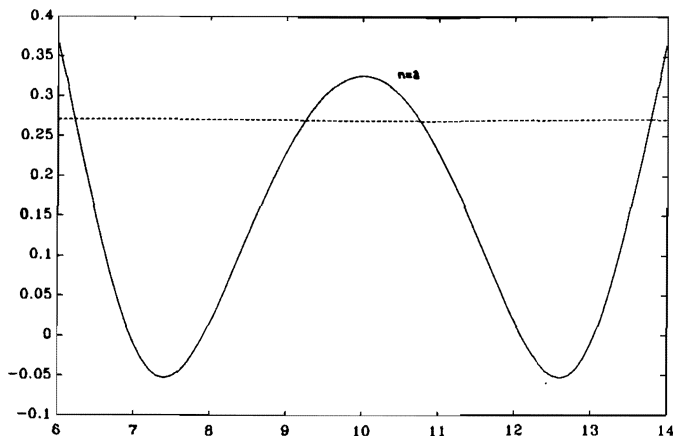
$$(4.2) \quad \pi_i(\bar{x}) = \frac{\pi_i f(\bar{x}|\theta_i)}{\sum_i \pi_i f(\bar{x}|\theta_i)}.$$

Hence

$$(4.3) \quad I_{\bar{x}} = \sum_i \pi_i(\bar{x}) \log \pi_i(\bar{x}) - \sum_i \pi_i \log(\pi_i).$$

In figure 6 $I_{\bar{x}}$ is plotted as a function of $\bar{x}$ for $n = 3$.

The amount of information for the normal prior is the constant line in the figure; it differs drastically from the amount of information from the data, using the histogram prior. A value of $\bar{x} = 6$ gives much more information for the prior in table 5 than for the normal because it reflects data that are in sharp contrast with the prior information. In fact, for priors such as in Table 5 with bounded parameter space, the information statistic is usually a *W*-shaped function of a sufficient statistic, centered over the marginal distribution of the sufficient statistic and the high information at extreme values is an indication of a possible error in either the prior or the data. A value of $\bar{x} = 7$, on the other hand, gives almost no information if the prior is as in table 5, since the data is just far enough away from the prior so that your knowledge of the parameter is not improved. This brings out clearly the point that the use of a normal approximation instead of the more specific prior can cause a wrong interpretation of the amount of knowledge obtained from the data.

Figure 6: Profile of $I_{\bar{x}}$ for the prior in table 5 when $n = 3$.**Example 3** (The binomial model)

For the binomial model

$$f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}, x = 0, 1, \dots, n$$

the Beta prior density,

$$\pi(\theta) = \frac{1}{\beta(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}, 0 < \theta < 1$$

is mostly used. As $\alpha \rightarrow \infty$, and $\beta \rightarrow \infty$, $I_x \rightarrow 0$. An interesting feature of the beta prior is that I_x is convex for all $\alpha, \beta > 0$ with a minimum when $x = \frac{n+\beta-\alpha}{2}$. Figure 7 shows the behaviour of I_x for $n = 10$ and for different values of $\alpha = \beta$.

The two vague priors normally used in the literature, $\alpha = \beta = \frac{1}{2}$ and $\alpha = \beta = 1$, give the maximum information from the sample with I_x for $\alpha = \beta = 1$ the larger when $2 < x < 8$. The expected information, $E_X(I_X)$, is however a maximum when $\alpha = \beta = \frac{1}{2}$, and decreases as α and β increases. This is of course Jeffreys' (1939/67) prior as derived by Bernardo (1979a). Assume now however, the following prior distribution on θ :

TABLE 6
PRIOR PROBABILITIES π_i FOR $\theta \in (0,08 - 0,92)$

θ	0,08 - 0,15	0,15 - 0,22	0,22 - 0,29	0,30 - 0,36	0,36 - 0,43
π_i	0,002	0,008	0,01	0,03	0,05
θ	0,43 - 0,5	0,5 - 0,57	0,57 - 0,64	0,64 - 0,71	0,71 - 0,78
π_i	0,4	0,4	0,05	0,03	0,01
θ	0,78 - 0,85	0,85 - 0,92			
π_i	0,008	0,002			

Taking $n = 10$, approximate values for I_x have been calculated using expressions similar to (4.2) and (4.3) and plotted in figure 8 together with the I_x values for the Beta prior with $\alpha = \beta = 17$, which has the same mean (0,5) and variance (0,007) as the distribution of table 6.

Figure 7: Profile of I_x for the binomial model with $n = 10$ and Beta prior with $\alpha = \beta = \frac{1}{2}, 1, 5, 50$.

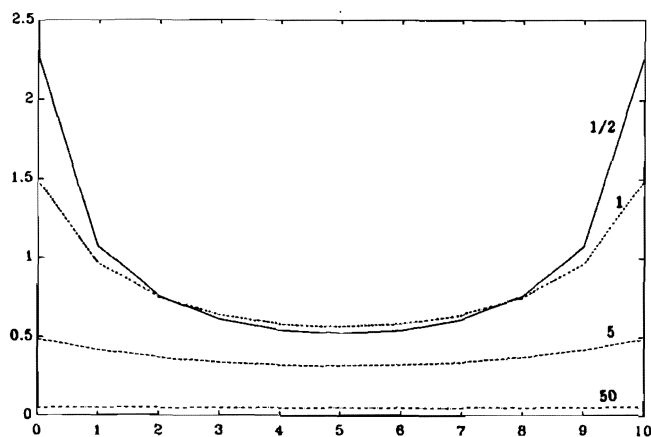
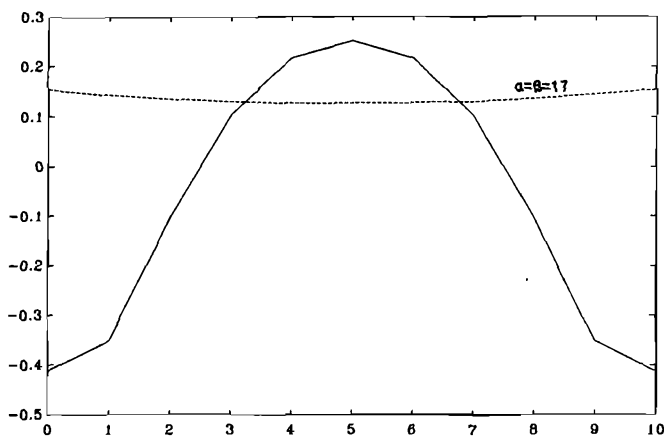


Figure 8: Profiles of I_x for prior in table 6 and I_x for Beta (17,17) prior.

Again the two profiles differ completely and, whereas extreme values of $x(0 \text{ or } n)$ give the most information with the Beta prior, the opposite occurs for the prior of table 6 which implies that the latter is a “stronger” prior in the sense that it is more reluctant to find a compromise with data that does not agree with it. The Beta prior, on the other hand, readily adapts to any sample information, as does the normal prior in example 2 and most smooth priors. The expected information from the data for both priors is however, about the same, namely 0,13.

In conclusion, these examples show that there can be a dramatic difference in the amount of information measured in a random sample, depending on whether the prior is reported in terms of a probability histogram (which is the easier and more natural for a non-statistician) or in terms of a smooth approximation such as a conjugate prior, which is often the standard procedure of the Bayesian analyst.

REFERENCES

ACZEL, J. and DAROCZY, Z. (1975). *On measures of information and their characterizations*. Academic Press.

BERGER, J.O. and O'HAGAN, A. (in press). Ranges of Posterior Probabilities for Unimodel Priors with Specified Quantiles, in *Bayesian Statistics III*, eds. J.M. Bernardo, M.H. DeGroot, D.V. Lindley and A.F.M. Smith. Oxford University Press, Oxford.

BERNARDO, J.M. (1979a). Reference posterior distributions for Bayesian inference. *J.R. Statist. Soc. B*, **41**, 113-147.

BERNARDO, J.M. (1979b). Expected Information as Expected Utility. *Ann. Statist.* **7**, 686-690.

BERNARDO, J.M. (1981). A Bayesian analysis of classical hypothesis testing. *Proceedings of first Valencia meeting on Bayesian Statistics*, University Press, Valencia, Spain.

DE GROOT, M.H. (1970). *Optimal Statistical Decisions*. McGraw-Hill, New York.

FEINSTEIN, A. (1958). *Foundations of Information Theory*. McGraw-Hille, New York.

GOOD, I.J. (1950). *Probability and the Weighing of Evidence*. Griffin: London.

GOOD, I.J. (1960). Weight of Evidence, Corroboration, Explanatory Power, Information and the Utility Experiments. *J.R. Statist. Soc. B*. **22**, 319-331.

HARTLEY, R.V. (1928). Transmission of information. *Bell System Tech. J.* **7**, 535-563.

JEFFREYS, H. (1939/1967). *Theory of Probability*. Oxford Clarendon Press.

KULLBACK, S. (1959). *Information Theory and Statistics*. Wiley, New York.

KULLBACK, S. and LEIBLER, R.A. (1951). Information and sufficiency. *Ann. Math. Statist.*, **22**, 79-86.

LINDLEY, D.V. (1956). On a measure of the information provided by an experiment. *Ann. Math. Statist.*, **27**, 986-1005.

LINDLEY, D.V. (1971). Bayesian Statistics, a Review. *Reg. Conf. Ser. Appl. Math.*, **2**. Philadelphia: SIAM.

MARTZ, H.F. and WALLER, R.A. (1982). *Bayesian reliability Analysis*. Wiley, New York.

OSTEYEE, D.B. and GOOD, I.J. (1974). *Information, Weight of Evidence, the Singularity between Probability Measures and Signal Detection*. Springer-Verlag, Berlin.

RAO, C.R. (1973). *Linear Statistical Inference and its Applications*. 2nd Ed. Wiley, New York.

RENYI, A. (1960). On measures of entropy and information. *Proc 4th Berkeley Symp. Math. Statist. Prob.* **1**, 547-561.

SHANNON, C.E. (1948). A mathematical theory of communication. *Bell System Tech. J.* **27**, 379-423, 623-656.